Technology Stack

Date	07 July 2026
Team ID	
Project Name	Personalized Networking Assistant
Maximum Marks	2 Marks

Define Your Technology Stack:

Identify the programming languages, frameworks, databases, front-end tools, back-end tools, and APIs that you will use to build your solution. A well-chosen technology stack ensures that your application is scalable, maintainable, and aligned with your project requirements.

Technology Stack Details

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	**Frontend / Client-Side**	Streamlit Framework (Python-based)	Enables fast layout rendering with native state configurations. Allowed building a clean, modern tab-based presentation interface without long frontend boilerplate code.

2	**Backend / Server-Side**	FastAPI	Built for high-speed routing execution using async Python practices. Automatically generates interactive API testing pages (Swagger UI) for error-free debugging.
3	**Database / Data Storage**	Local JSON Flat-Files (history.json & feedback.json)	Lightweight data handling mechanism. Since the system requires lightweight local interaction storage logs, structured database instances are redundant.
4	**Cloud / Hosting / Deployment**	Virtual Environment Sandbox (Python 3.11+)	Isolates development dependencies cleanly, preventing resource version mismatch conflicts across different developer setups.
5	**Version Control & CI/CD**	Git & GitHub	Facilitates source control management. Allowed the group to review commits, handle parallel updates efficiently, and maintain clean deploy snapshots.
6	**Third-Party APIs / Other Tools**	Hugging Face Transformers (DistilBERT & GPT-2), python-wikipedia-api	Hugging Face pipelines facilitate access to zero-shot classifiers and text generators. Wikipedia API grants direct access to reference lookup definitions.